

Md Khorshed Alam

Curriculum Vitae

Profile

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Nationality: Bangladeshi

Computer Science researcher and university lecturer with thesis-based M.Sc. training in geometric deep learning, a Q1 journal article, two Springer proceedings papers, three manuscripts under 1st or 2nd review, a competitive national research fellowship, and more than five years of university teaching experience. My proposed doctoral research, developed in consultation with a confirmed AUT-based supervisory team, addresses reliability-centred decision-making for vision–language–action (VLA) embodied agents; moving beyond whether a model can propose a plausible action toward whether the surrounding system can ground that action in physical constraints, predict its feasibility, recall causally relevant past failures, and arbitrate safety before execution.

Research Interests Embodied AI; Vision–Language–Action Models; Long-Horizon Planning under Uncertainty; Feasibility and World-Model Prediction; Causal Memory; Safety Arbitration; Cognitive Architectures for Reliable Autonomy.

Proposed Doctoral Research

Working Title *Toward Cognitive Architectures for Reliable Long-Horizon Planning in LLM-Based Embodied Agents*. Proposes a Cognitive Reliability Loop (CRL); a modular decision layer combining constraint-aware planning, feasibility prediction, causal memory, and safety arbitration; around a VLA policy (OpenVLA), evaluated first in the LIBERO simulation benchmark and staged toward Unitree G1 humanoid deployment.

Confirmed Supervisory Team

- **Dr. Trung Nguyen** (Primary Supervisor) — Lecturer, Department of Computer and Information Sciences, Auckland University of Technology.
- **Dr. Abubakar Siddique** (Co-Supervisor) — Lecturer, Department of Electrical & Electronic Engineering, Auckland University of Technology; research spans explainable AI, evolutionary computing, and cognitive architectures.
- **Dr. M. A. Hakim Newton** (International Co-Supervisor) — Senior Lecturer in Data Science, School of Information and Physical Sciences, The University of Newcastle, Australia; research spans artificial intelligence, intelligent search, and optimization.

Education

2021–2025 **M.Sc. in Computer Science & Engineering**, Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh, **CGPA: 3.25/4.00**

- **Thesis:** Enhancing Protein–Ligand Pose Prediction via Deep Learning with Transformation-Invariant Loss Function and Iterative Refinement — independent research project including formal research-methodology coursework, culminating in three peer-reviewed/under-review manuscripts.
- **Supervisor:** Dr. Mohammed Eunus Ali, former Professor, Department of CSE, BUET; currently Senior Lecturer, Department of Software Systems & Cybersecurity, Monash University.
- **Collaborator:** Dr. M. A. Hakim Newton, University of Newcastle, Australia.
- **Institutional Ranking:** QS World University Rankings 2026 (BUET): #301–350 in Electrical & Electronic Engineering; #451–500 in Computer Science & Information Systems. THE World University Rankings 2026 (Computer Science): #601–800.

2014–2018 **B.Sc. in Computer Science & Engineering**, Chittagong University of Engineering and Technology (CUET), Chattogram, Bangladesh, **CGPA: 3.48/4.00**

- **Institutional Ranking:** THE World University Rankings 2026 (Computer Science): #601–800.

English Proficiency

IELTS Academic, overall 7.0 (Listening 7.5, Reading 7.5, Writing 7.5, Speaking 6.0), 19 September 2025.

Funding, Awards and Honours

Research Fellowship

ICT Research Fellowship, Government of Bangladesh (2021–2023) — competitive national award for M.Sc. research on deep-learning-based protein–ligand docking and pose refinement.

Academic Scholarships

Government Talent Scholarships, Bangladesh — awarded in Year 5, Year 8, and Higher Secondary for academic excellence (top-percentile, nationwide merit list).

Peer-Reviewed Publications

2026 Chowdhury, A. P., Nur, F. N., Islam, A. H. M. S., **Alam, K.**, Karim, A., & Shah, M. A. **FLEX-IDS: A Secure and Explainable Federated Intrusion Detection Framework for Edge-of-Things Environments under Adversarial Conditions**. *Computers & Electrical Engineering*, 129(B), 110827. **SCIImago Q1; Legacy CORE2020 Rank B.**

- 2023 **Alam, M. K., & Saha, T. Secrecy Rate Optimization for Energy Efficient Cognitive Relay Networks.** In *Innovations in Computer Science and Engineering*, Lecture Notes in Networks and Systems, Vol. 565, Springer.
- 2023 **Saha, T., & Alam, M. K. Comparison Between Property-Based Software Security Testing Technique and Fault Injection.** In *Innovations in Computer Science and Engineering*, Lecture Notes in Networks and Systems, Vol. 565, Springer.
- 2026 **Thesis-derived manuscript on protein–ligand pose prediction and iterative geometric refinement.** Submitted to *Applied Intelligence*; under second-round review.
- 2026 **GeoProNet: Geometric Graph Transformer for Protein–Ligand Pose Refinement with Torsion-Informed Learning.** Submitted to *Applied Intelligence*; under first-round review.
- 2026 **GeoProNetUp: SE(3)-Equivariant Geometric Graph Refinement for Reliable Protein–Ligand Pose Correction in Medical Informatics Pipelines.** Submitted to *Applied Intelligence*; under first-round review.

Research Experience

- 2023–Present **Master’s Thesis Researcher, BUET, Dhaka, Bangladesh**
- Designed transformation-invariant geometric deep-learning pipelines for protein–ligand pose correction and developed iterative, reliability-focused refinement models.
 - Implemented reproducible training and evaluation workflows in Python using PyTorch, producing three thesis-derived journal manuscripts.
- 2021–2023 **Research Assistant, BUET, Dhaka, Bangladesh**
- Investigated ML-based protein–ligand docking and structural/chemical feature representations.
 - Analysed CPython garbage collection and accelerated a Conjugate Gradient Method implementation.

Academic and Professional Experience

- 2024–Present **Lecturer, Department of CSE, Notre Dame University Bangladesh, Dhaka, Bangladesh**
- Teach undergraduate CSE courses (incl. Artificial Intelligence) and mentor students in programming and research preparation.
 - Contribute to OBE curriculum development, accreditation documentation, and academic quality assurance.
- 2019–2023 **Lecturer, Department of CSE, City University; State University of Bangladesh, Dhaka, Bangladesh**
- Taught core CSE courses; contributed to OBE-based syllabus development and research-methodology seminars; supported CSE Fest and inter-university programming-contest activities.

Leadership and Service

- Convener, TP-Link | Excel Presents NDUB CSE Fest 2026** — coordinated a three-day national event involving 86 teams from more than 61 universities.
- Initiated competitive-programming coaching at Notre Dame University Bangladesh; invited speaker, departmental seminar, 2025.
- President, Nobojagoron Torun Songho** — led educational, library-development, environmental, flood-relief, and youth-development initiatives.

Technical Skills

- Programming & ML Python, C/C++, Java, MATLAB; PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
- Scientific Tools RDKit, Open Babel, PyMol, AutoDock Tools
- Research Tools Git, L^AT_EX, Linux, Jupyter, remote GPU servers
- Development Training in Outcome-Based Education and curriculum development; Workshop on Machine Learning for Drug Discovery, 2023

Academic Referees

Name	Position and Affiliation	Email
Dr. Mohammed Eunus Ali	Senior Lecturer, Department of Software Systems and Cybersecurity, Monash University	Eunus.Ali@monash.edu
Dr. M. A. Hakim Newton	Lecturer, University of Newcastle, Australia (also confirmed International Co-Supervisor for the proposed AUT PhD)	Mahakim.Newton@newcastle.edu.au
Dr. Asaduzzaman	Professor, Department of CSE, CUET	asad@cuet.ac.bd